

COMPUTER VISION – CASE STUDY

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1. A remote sensing application is used to align satellite images of the same geographical region captured at different times for monitoring environmental changes. The system initially relies on edge detection to identify matching regions between the images; however, due to differences in illumination, image orientation, and background details, the matching process becomes unreliable and produces inaccurate alignment results. The development team decides to replace the edge-based approach with Harris Corner Detection to identify stable and distinctive feature points. Analyze the limitations of edge detection in this application and explain how Harris Corner Detection improves feature localization, matching accuracy, and robustness for image registration under varying imaging conditions.

ANSWER:

A remote sensing system must align satellite images of the same geographical region captured at different times. The original edge-based approach becomes unreliable because illumination, orientation and background details vary. The proposed solution is to use Harris Corner Detection to identify stable and distinctive feature points.

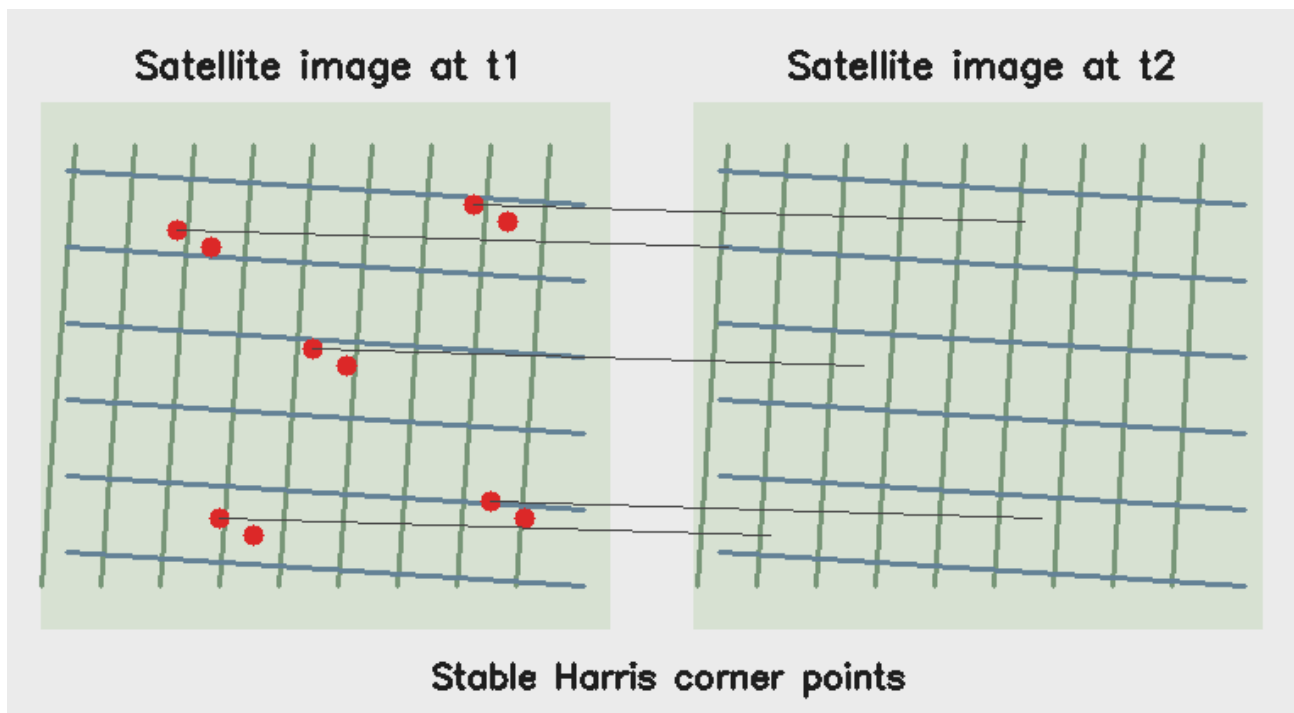


Figure 1. Illustrative satellite-image registration using stable corner points.

1. Understanding of the Case – 25%

The key problem is inaccurate image registration when the same geographical region is observed under different imaging conditions. Edge detection mainly identifies intensity transitions and may produce long, fragmented or ambiguous edges. Changes in illumination can alter edge strength, while image orientation and background structures can cause corresponding edges to be difficult to identify. Therefore, edge features alone are not sufficiently distinctive for reliable matching.

2. Application of Concepts – 25%

Harris Corner Detection is more suitable because it searches for points where image intensity changes significantly in different directions. A corner is therefore more localized than a general edge and can act as a distinctive reference point. The Harris response is based on the local gradient structure and identifies locations with strong two-dimensional intensity variation.

- Preprocess the satellite images to reduce noise and normalize the images when required.
- Compute image gradients in the horizontal and vertical directions.
- Construct the local second-moment matrix from the gradient information.
- Calculate the Harris response and select strong local maxima as corner points.
- Match corresponding corner points between the two satellite images and estimate the image transformation for registration.

3. Analysis – 20%

Edge detection is sensitive to illumination and often gives features that are not unique. An edge may continue across a large region, making correspondence ambiguous. Harris corners are spatially localized and contain intensity variation in multiple directions, so they provide stronger geometric references for alignment. This improves feature localization and makes matching more reliable when the image content changes moderately.

4. Solution Design – 20%

A feasible registration pipeline is: acquire the two satellite images → preprocess and normalize → detect Harris corners in both images → compare corresponding feature neighborhoods → reject incorrect matches → estimate the geometric transformation from reliable correspondences → warp one image to the coordinate system of the other → verify alignment. This design directly addresses the weaknesses of the original edge-based approach.

5. Clarity – 10%

Overall, Harris Corner Detection provides stable, localized and distinctive feature points, leading to improved localization, matching accuracy and robustness for satellite image registration under varying imaging conditions.

2. An autonomous surveillance system continuously captures high-resolution color images to detect vehicles and pedestrians in real time. Processing the original color images directly increases computational complexity and slows down feature extraction, making it difficult to achieve the required processing speed. To improve system performance, the engineers consider converting the images into grayscale or binary representations before applying feature detection algorithms. Analyze the impact of different image representations on computational efficiency, edge detection, and feature extraction performance, and justify the most appropriate image representation for achieving accurate and reliable object detection in real-time computer vision applications.

ANSWER:

An autonomous surveillance system processes high-resolution color images to detect vehicles and pedestrians in real time. Direct processing of RGB images increases computational cost. The case therefore compares grayscale and binary representations for feature detection and extraction.



Figure 2. Web-searched example of RGB/RGBA, grayscale and black-and-white image representations used to illustrate the case.

1. Understanding of the Case – 25%

The main requirement is accurate object detection with sufficiently low processing time. RGB images contain three color channels, so feature algorithms must process more data. Grayscale represents each pixel with a single intensity value, while binary images reduce each pixel to two levels. The choice affects computational efficiency, edge quality and the information available for feature extraction.

2. Application of Concepts – 25%

RGB representation preserves color information and can be useful when color is important for distinguishing objects, but it requires greater computation. Grayscale removes color information while retaining intensity structure, making it suitable for many edge and feature-detection algorithms. Binary representation is even simpler, but thresholding can remove useful object details and make detection sensitive to lighting and threshold selection.

- RGB: highest information content and computational requirement.
- Grayscale: one intensity channel, lower computation, and good support for edges and corners.
- Binary: very low computational cost, but may lose texture, shading and fine object information.
- For real-time detection, grayscale is generally the best balance when color is not essential to the detection task.

3. Analysis – 20%

For edge detection, grayscale is particularly effective because common edge operators use intensity gradients. It reduces the data volume from three channels to one while preserving the intensity changes required to locate object boundaries. Binary images can produce strong boundaries after thresholding, but they may discard weak edges and internal features. Consequently, binary processing may be faster but less reliable in scenes with shadows, illumination changes or complex backgrounds.

4. Solution Design – 20%

The recommended real-time pipeline is: capture RGB image → convert to grayscale → optionally denoise or normalize → perform edge/corner or other feature detection → extract features → detect and classify vehicles and pedestrians. Binary conversion can be added selectively when a stable threshold or segmentation stage is already available. This approach reduces computation while retaining the intensity information needed for reliable feature extraction.

5. Clarity – 10%

Therefore, grayscale is the most appropriate default representation for accurate and reliable real-time object detection when color information is not required. It offers a practical compromise between computational efficiency and preservation of useful visual features.